

TurboPress: A Measurement-Driven Study of Rotated Low-Bit Weight Quantization — Equilibration Exponents, Error Feedback, and Per-Layer Bit Allocation

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Abstract

We present **TurboPress**, an instrumented post-training quantization (PTQ) pipeline for large language model (LLM) weights, together with a set of controlled A/B measurements that isolate the contribution of each design choice at *exactly matched* bit budgets. Building on the incoherence-processing line of work [2, 3, 17, 18] and on the rotation+optimal-quantizer construction of TurboQuant [23], TurboPress combines a seeded randomized orthogonal rotation, trellis-coded quantization (TCQ) with data-free trellis-optimized codebooks, GPTQ/LDLQ error feedback [3, 5], and per-layer bit allocation. Our contributions are primarily *empirical and analytical* rather than a claim of a new state of the art: (i) a corrected rotation-aware *equilibration exponent*—we show that in the weight-only, post-rotation regime the optimal per-channel fold is a quarter power $s_j = (\mathbb{E}[x_j^2]/\|\mathbf{W}_{:,j}\|^2)^{1/4}$, distinct from the square-root heuristic of AWQ/SmoothQuant [9, 21] and from the weight–activation balancing result of BASE-Q [6]; (ii) a rigorous *negative result* that unbiased (QJL-style) residual correction, while provably debiasing, loses to a finer deterministic quantizer at equal budget for weights; and (iii) an end-to-end measurement on Qwen3-0.6B showing that error feedback and trellis coding are complementary (trellis wins at 2 bits; both are near-lossless at 4 bits), and that KL-sensitivity-driven mixed precision beats uniform precision at an equal 3.0-bit budget. All numbers are reported with a KL divergence “visible-loss” metric and exact storage accounting; code and configuration are released.

1 Introduction

Low-bit weight quantization is the dominant lever for shrinking LLM memory and bandwidth. The strongest modern PTQ methods share a common backbone: *incoherence processing* (a random or learned rotation) to spread outliers [2, 3, 10], an information-theoretically strong quantizer (vector or trellis codebooks) [17, 18, 20], and Hessian-aware *error feedback* [4, 5]. TurboQuant [23] adds a Quantized Johnson–Lindenstrauss (QJL) sketch [22] to make inner-product estimates *unbiased*.

This paper does not propose a new frontier method. Instead we ask a narrower, measurable question: *at matched bit budgets, how much does each component actually buy, and why?* We build TurboPress as a harness in which rotation, quantizer, equilibration, error feedback, and bit allocation can each be toggled and measured against a full-precision reference using KL divergence $\text{KL}(p_{\text{fp}}\|p_q)$, top-1 next-token agreement, and perplexity, with storage reported to the exact fractional bit.

Contributions.

- **Corrected equilibration exponent (§3.1).** Under a rotation-based quantizer whose error is isotropic in the rotated basis, the expected output error factorizes as $\left(\sum_j c_j s_j^2\right)\left(\sum_j m_j / s_j^2\right)$ with $c_j = \|\mathbf{W}_{:,j}\|^2$ and $m_j = \mathbb{E}[x_j^2]$; Cauchy–Schwarz gives the optimum $s_j = (m_j / c_j)^{1/4}$. This is a *quarter* power, not the square root used by AWQ/SmoothQuant, and it arises from a different objective than the weight–activation balance that yields the 1/2 power in BASE-Q.
- **Error feedback in rotated coordinates (§3.3).** We implement GPTQ/LDLQ in the rotated, equilibrated basis via a fast transform of the activation Hessian, and show on Qwen3-0.6B that it reduces 2-bit KL by 5.0× and 3-bit KL by 5.3× over nearest rounding, at $\sim 3\times$ lower quantization cost than TCQ.
- **Per-layer bit allocation (§3.4).** A measured KL-sensitivity probe plus a greedy budget knapsack beats uniform precision at an *equal* 3.0-bit budget (KL 0.238 vs. 0.276).
- **A refuted hypothesis (§3.5).** Unbiased QJL residual correction for weights is measured to lose to deterministic quantization at equal budget, in both synthetic depth-compounding chains and on a real model; the correction’s variance exceeds the bias it removes.

2 Background and Related Work

Rotation / incoherence processing. QuIP [3] showed that pre- and post-multiplying weights by random orthogonal matrices makes them incoherent, tightening rounding guarantees; QuIP# [17], QuaRot [2], and SpinQuant [10] refine this with Hadamard and learned rotations. A random rotation renders the rotated coordinates approximately Gaussian, which is the regime where the Lloyd–Max scalar quantizer [11, 13] is optimal.

Quantizers. Beyond scalar rounding, vector and trellis codebooks close much of the gap to the rate–distortion bound. Trellis-coded quantization (TCQ) [12], built on Ungerboeck set partitioning [19], was recently applied to LLM weights by QTIP [18]; Pyramid Vector Quantization is another route [20].

Error feedback. OBQ/GPTQ [4, 5] quantize input channels sequentially and propagate rounding error through the inverse Hessian; LDLQ [3] is the equivalent within incoherence processing. Qronos [24] generalizes the correction step.

Activation-aware scaling / equilibration. Weight equalization [14], SmoothQuant [21], and AWQ [9] rescale channels by (a power of) their activation magnitude; AWQ grid-searches the exponent $\alpha \in [0.25, 0.75]$. BASE-Q [6] derives an optimal scale via AM–GM for the *weight-and-activation* setting, obtaining a square-root form. Recent concurrent work combines several of these ingredients: QAM-W [16] pairs Hadamard rotation with a grid-tuned AWQ scale and a 2D codebook; OrbitQuant [8] uses a rotated basis with a Lloyd–Max codebook and no per-channel scaling; PiSO [1] computes exact per-channel scales algorithmically. We position our equilibration result precisely against these in §3.1 and §6.

3 Method

Let a linear layer compute $\mathbf{y} = \mathbf{W}\mathbf{x}$ with $\mathbf{W} \in \mathbb{R}^{m \times d}$. TurboPress replaces $\mathbf{W}$ by a quantized $\hat{\mathbf{W}}$ acting on a transformed input, exactly factoring the computation as

$$\mathbf{y} = \mathbf{W}\mathbf{x} = (\mathbf{W}\mathbf{D}\mathbf{R}^\top)(\mathbf{R}\mathbf{D}^{-1}\mathbf{x}), \quad \mathbf{z} := \mathbf{R}\mathbf{D}^{-1}\mathbf{x}, \quad (1)$$

where $\mathbf{D} = \text{diag}(\mathbf{s})$ is an activation-aware equilibration and $\mathbf{R}$ is a seeded randomized orthogonal transform (block Walsh–Hadamard $\times$ Rademacher signs, with an exact dense fallback for odd dimensions). We quantize $\mathbf{W}'' = \mathbf{W}\mathbf{D}\mathbf{R}^\top$ and run $\mathbf{y} \approx \hat{\mathbf{W}}''\mathbf{z} + \mathbf{b}$. All of $\mathbf{R}$, $\mathbf{D}$, and the codebook are seeded and reproducible; only the low-bit codes and per-row scales are stored.

3.1 Rotation-aware equilibration: the quarter-power rule

The rotation spreads the per-row quantization error approximately isotropically across the d rotated coordinates. We model the error $\mathbf{E}'' = \mathbf{W}'' - \hat{\mathbf{W}}''$ in the equilibrated basis as column-decorrelated with per-entry second moment proportional to the row energy: $\mathbb{E}[(\mathbf{W}' - \hat{\mathbf{W}}')_{ij}^2] \approx \delta_i^2$ with $\delta_i^2 = (\varepsilon^2/d)\|\mathbf{W}_{i,:}'\|^2 = (\varepsilon^2/d)\sum_j s_j^2 W_{ij}^2$, where ε^2 is the normalized quantizer MSE and $\mathbf{W}' = \mathbf{W}\mathbf{D}$.

Proposition 1 (Optimal equilibration under isotropic rotated error). *With $c_j = \|\mathbf{W}_{:,j}\|^2$ and $m_j = \mathbb{E}[x_j^2]$, the expected squared output error satisfies*

$$\mathbb{E}\|\Delta\mathbf{y}\|^2 \approx \frac{\varepsilon^2}{d} \left(\sum_j c_j s_j^2 \right) \left(\sum_j m_j / s_j^2 \right), \quad (2)$$

which, by the Cauchy–Schwarz inequality, is minimized when $c_j s_j^2 \propto m_j / s_j^2$, i.e.

$$s_j = (m_j / c_j)^{1/4} = (\sqrt{\mathbb{E}[x_j^2]} / \|\mathbf{W}_{:,j}\|)^{1/2}. \quad (3)$$

Sketch. The output error in row i is $\Delta y_i = \sum_j (\mathbf{W}' - \hat{\mathbf{W}}')_{ij} x_j / s_j$. Column decorrelation of the rotated error kills cross terms in expectation, giving $\mathbb{E}[\Delta y_i^2] \approx \delta_i^2 \sum_j m_j / s_j^2$. Summing over i and using $\sum_i \delta_i^2 = (\varepsilon^2/d) \sum_j s_j^2 \sum_i W_{ij}^2 = (\varepsilon^2/d) \sum_j c_j s_j^2$ yields (2). Cauchy–Schwarz on the two sums gives equality when the summands are proportional, which is (3). $\square$

Interpretation. The AWQ/SmoothQuant fold $s_j \propto \mathbb{E}[x_j^2]^{1/2}$ is optimal only when the quantizer error stays *column-aligned* (its variance in column j scales as $s_j^2 W_{ij}^2$). The random rotation destroys that alignment—the error becomes isotropic across rotated coordinates—so the two competing terms in (2) both scale with the weight column, and the optimum shifts to a quarter power. This is a distinct regime from BASE-Q [6], whose 1/2 exponent balances *activation* quantization noise against *weight* quantization noise (both quantized); here activations are not quantized. The prediction $\alpha \approx 1/4$ is consistent with the empirically favored small AWQ exponents ($\alpha \in [0.25, 0.75]$, 9) and the frozen $\alpha=0.3$ of QAM-W [16]. We treat this as a narrow analytical result under stated modeling assumptions, not a claim of a new method (§6).

3.2 Data-free trellis-optimized codebooks

We use rate- R TCQ with a doubled (2^{R+1} -level) codebook and a rate-1/2 convolutional trellis over S states; Viterbi decoding selects the minimum-distortion codeword *sequence* per row, storing exactly R bits/weight (1 path bit + $R-1$ member bits, verified by an encode→decode round-trip

test). Because the rotation makes the coordinate distribution known ($\approx \mathcal{N}(0, 1)$), we re-optimize the codebook *under the trellis* with generalized Lloyd iterations on synthetic Gaussians—no calibration data—and cache the result.

3.3 GPTQ/LDLQ error feedback in rotated coordinates

Nearest rounding leaves each layer’s output error as a fixed bias. LDLQ/GPTQ [3, 5] instead quantize input channels sequentially and feed each channel’s rounding error forward into the not-yet-quantized channels, weighted by the layer’s input Hessian $\mathbf{H} = \mathbb{E}[\mathbf{x}\mathbf{x}^\top]$, minimizing $\text{tr}[(\mathbf{W}'' - \hat{\mathbf{W}}'')^\top \mathbf{H}_z (\mathbf{W}'' - \hat{\mathbf{W}}'')]^\top$ rather than raw weight MSE. The relevant Hessian in the transformed space of (1) is $\mathbf{H}_z = \mathbf{R}\mathbf{D}^{-1}\mathbf{H}\mathbf{D}^{-1}\mathbf{R}^\top$, which we form with two applications of the fast transform ($O(d^2 \log d)$). The random rotation is exactly the incoherence preprocessing that makes LDLQ effective. Storage and bit accounting are identical to the plain scalar path; only the chosen codeword changes. When $\mathbf{H}_z$ is diagonal, LDLQ provably reduces to independent nearest rounding (a unit test verifies bit-exact equivalence).

3.4 Per-layer bit allocation by measured KL sensitivity

Layers differ in sensitivity, so a uniform width wastes budget. For each quantized linear and each candidate width $b \in \{2, 3, 4\}$ we quantize *only* that layer (others full precision) and measure $\text{KL}(p_{\text{fp}} \| p_q)$ on a held-out probe set—a direct, model-faithful sensitivity signal. We then choose a width per layer to minimize total predicted KL subject to a parameter-weighted average-bit budget, solved greedily by repeatedly spending the next unit of budget on the upgrade with the largest KL drop per added bit. Greedy is exact here because per-layer KL is empirically convex-decreasing in bits.

3.5 A refuted hypothesis: unbiased (QJL) correction for weights

TurboQuant’s QJL sketch [22, 23] makes a layer output an *unbiased* estimate of the full-precision output. The hypothesis we test: since deterministic rounding produces a fixed per-layer bias that could compound through depth, spending bits on an unbiasing sketch might beat spending them on a finer deterministic quantizer. We implement the 1-bit QJL residual sketch (with the exact unbiasedness and variance bound $\text{Var}[\langle \hat{r}, x \rangle] \leq (\pi/2) \|r\|^2 \|x\|^2 / k$) and measure it at equal budget.

4 Experimental Setup

We quantize every `nn.Linear` in the decoder blocks of a Hugging Face causal LM (embeddings, LM head, and norms kept in full precision) and compare to the unquantized model on real text (a fixed public-domain corpus). Primary model: Qwen3-0.6B [15] (28 blocks, 196 linears) in fp16 on a single 8 GB GPU; scale-up reference: SmolLM2-135M [7]. Metrics on 4,080 next-token positions: mean $\text{KL}(p_{\text{fp}} \| p_q)$, top-1 agreement, and perplexity; equilibration scales and Hessians are calibrated on a disjoint 2,048-token slice. Bits/weight is the exact storage-weighted average including per-row scales and codebooks. Everything is seeded and deterministic on a given device.

5 Results

5.1 Rate-distortion on Qwen3-0.6B

Table 1 and Fig. 1 give the full sweep (fp16 perplexity 29.11). Nearest rounding collapses at 2 bits (perplexity 5.6×10^5); both trellis coding and error feedback rescue it, and at 4 bits both are within $\sim 5\%$ perplexity of full precision with $\text{KL} \leq 0.10$.

Table 1: Qwen3-0.6B, all 196 decoder linears quantized; fp16 perplexity 29.11. “+eq” is the quarter-power equilibration of (3). Best per bit in **bold**. Quantization wall-clock on one 8 GB GPU.

Configuration	bits/w	$\text{KL}(p_{\text{fp}} \ p_q)$	top-1	perplexity	time (s)
Nearest (rotated scalar) 2b	2.012	10.282	0.020	564,874	4.6
Nearest 3b	3.013	1.641	0.381	138.66	4.4
Nearest 4b	4.013	0.399	0.670	43.91	4.0
TCQ($S=64$)+eq 2b	2.023	1.215	0.450	91.42	245.9
TCQ($S=64$)+eq 3b	3.023	0.289	0.710	38.58	245.2
TCQ($S=64$)+eq 4b	4.023	0.079	0.843	30.63	247.8
GPTQ/LDLQ+eq 2b	2.023	2.072	0.383	177.10	79.0
GPTQ/LDLQ+eq 3b	3.023	0.312	0.707	36.28	88.8
GPTQ/LDLQ+eq 4b	4.023	0.095	0.844	30.45	77.2

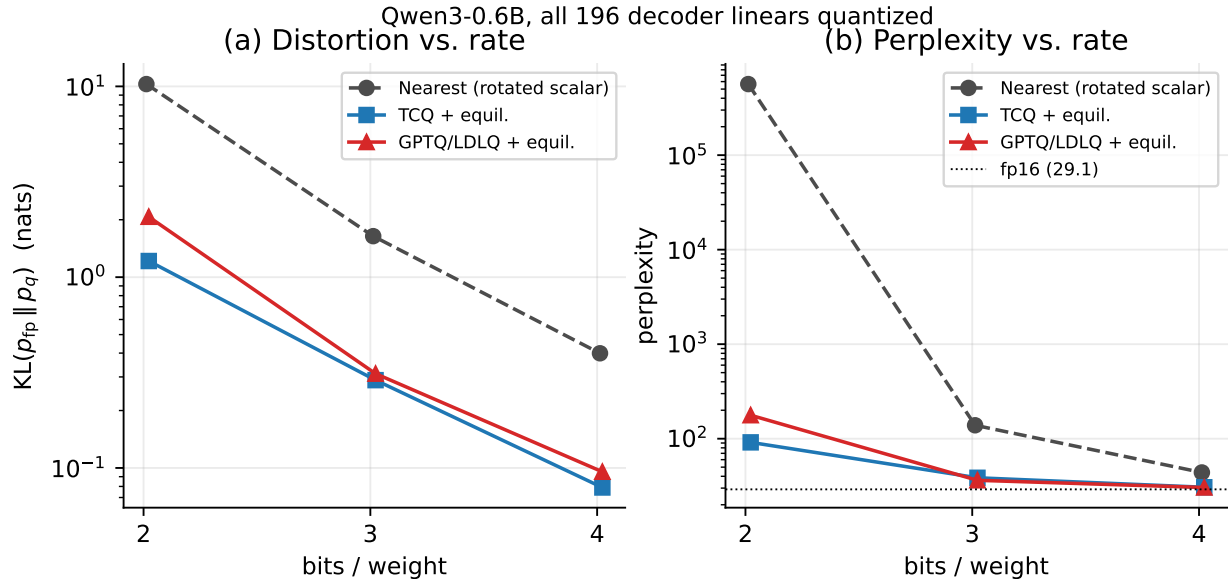


Figure 1: Rate-distortion on Qwen3-0.6B. Both trellis coding and error feedback dominate nearest rounding across the whole range; they are near-lossless at 4 bits.

5.2 Error feedback vs. trellis coding: complementary

Fig. 2a isolates the KL of each family. Error feedback cuts 2-bit KL $10.28 \rightarrow 2.07$ ($5.0\times$) and 3-bit KL $1.64 \rightarrow 0.31$ ($5.3\times$) over nearest, at $\sim 3\times$ lower cost than TCQ (79s vs. 246s). GPTQ/LDLQ 3b beats *nearest* 4b (KL 0.312 vs. 0.399)—a full bit saved. The two strong methods tie at 3–4 bits,

but the joint trellis code wins clearly at the hardest 2-bit setting (KL 1.22 vs. 2.07), where scalar codebook granularity is coarsest—motivating combining them (§6).

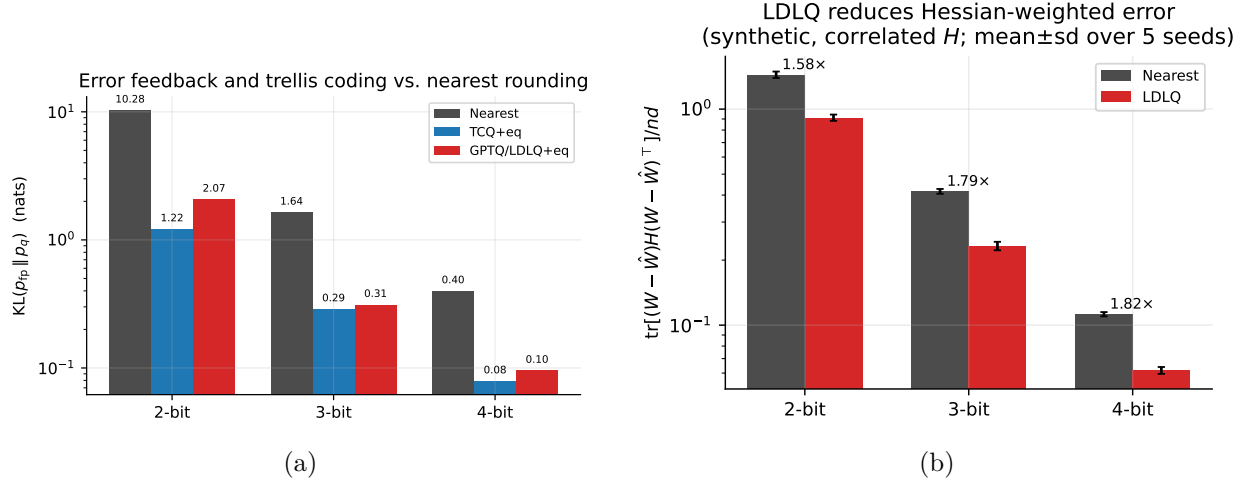


Figure 2: (a) KL by method and bit-width on Qwen3-0.6B (log scale). (b) Controlled microbenchmark: LDLQ reduces the Hessian-weighted output error $\text{tr}[(\mathbf{W} - \hat{\mathbf{W}})\mathbf{H}(\mathbf{W} - \hat{\mathbf{W}})^\top]/nd$ by 1.6–1.8× over nearest on a synthetic layer with correlated $\mathbf{H}$.

5.3 The equilibration exponent matters: quarter power beats square root

Fig. 3 tests Proposition 1 directly: an identical rotated scalar quantizer in which *only* the activation fold changes. On Qwen3-0.6B the quarter-power fold cuts 2-bit KL from 10.28 (no equilibration) to 2.40, where the square-root fold reaches only 8.41. On Qwen3-1.7B the square-root fold is *worse than no equilibration at all* at 3 and 4 bits (KL 3.58 vs. 2.24, and 1.85 vs. 0.64), while the quarter fold improves every setting (0.46, 0.18). This is the failure mode (3) predicts: over-scaling by $\sqrt{m_j}$ inflates the folded weight energy $\sum_j c_j s_j^2$ faster than it suppresses the activation term, and the imbalance grows with the channel spread of the model.

5.4 Scale-up sharpens the low-bit picture

Fig. 4 compares SmolLM2-135M to Qwen3-0.6B for the same TCQ+eq method. The 2-bit perplexity ratio $\text{ppl}_q/\text{ppl}_{\text{fp}}$ drops from $8.3 \times (116.1/14.06)$ to $3.1 \times (91.4/29.11)$; the 4-bit perplexity gap shrinks from +8.1% to +5.2%, while 4-bit KL is essentially size-invariant (0.0787 vs. 0.0790). Larger models quantize better at the aggressive settings, consistent with the broader literature.

5.5 Per-layer bit allocation

The sensitivity map (Fig. 5) shows MLP projections dominate (mean single-layer 2-bit KL: down 0.0088, up 0.0080, gate 0.0062), then attention output/value (0.0047), with query/key nearly free (0.0025/0.0016); block 0 and the final block are hotspots. Allocating against this signal (Table 2, Fig. 6) beats uniform precision at an *equal* 3.0-bit budget—KL 0.238 vs. 0.276 (−14%), perplexity 34.76 vs. 36.59—by moving 24 insensitive layers to 2 bits and 23 sensitive ones to 4 bits. At a fractional 2.5-bit budget (impossible for a uniform integer width) the mixed model reaches KL 0.494, roughly one third of the way from uniform-2b (1.386) to uniform-3b.

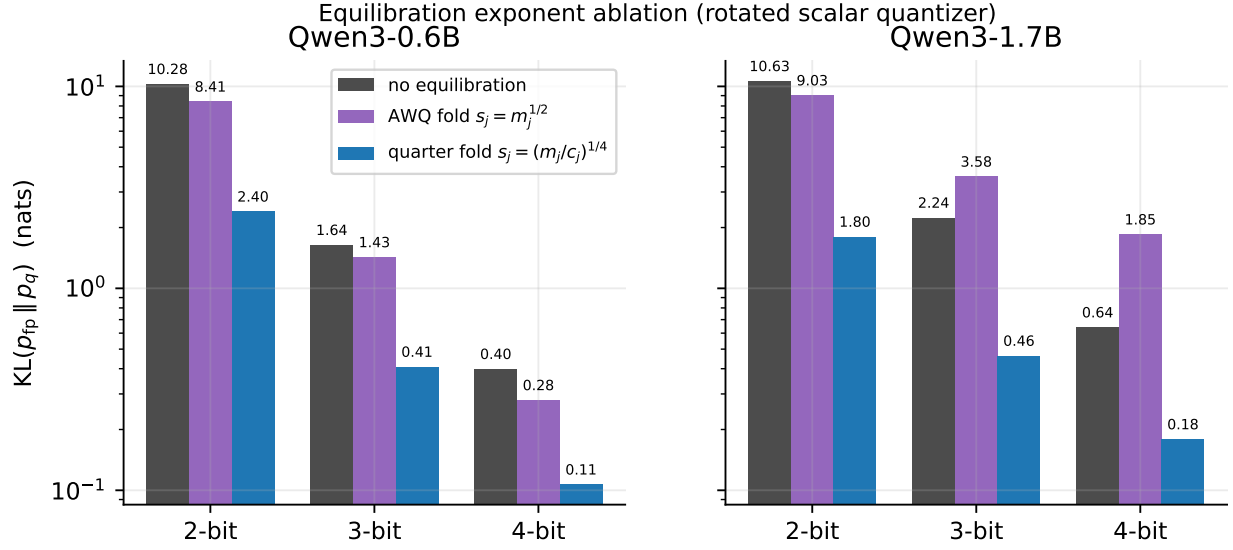


Figure 3: Equilibration-exponent ablation (Proposition 1): identical rotated scalar quantizer, only the per-channel fold differs. The quarter-power fold $s_j = (m_j/c_j)^{1/4}$ dominates at every bit-width on both models; on Qwen3-1.7B the AWQ-style square-root fold is worse than no equilibration at 3–4 bits. Log scale; calibration and evaluation slices are disjoint.

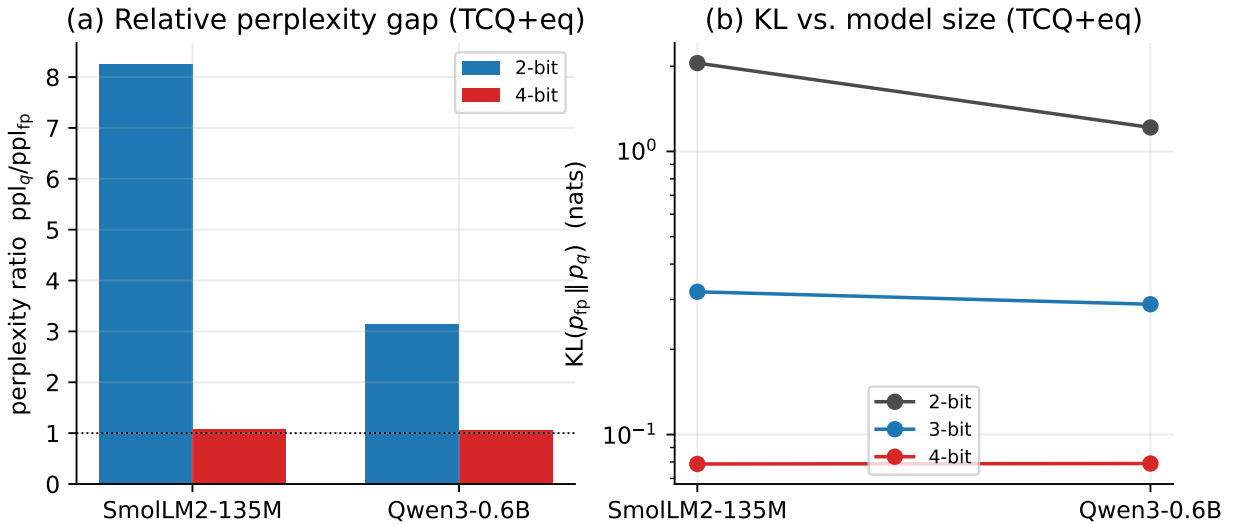


Figure 4: Scale-up from 135M to 0.6B parameters (TCQ+eq). (a) Relative perplexity gap; (b) KL by width. The 2-bit setting improves most with scale.

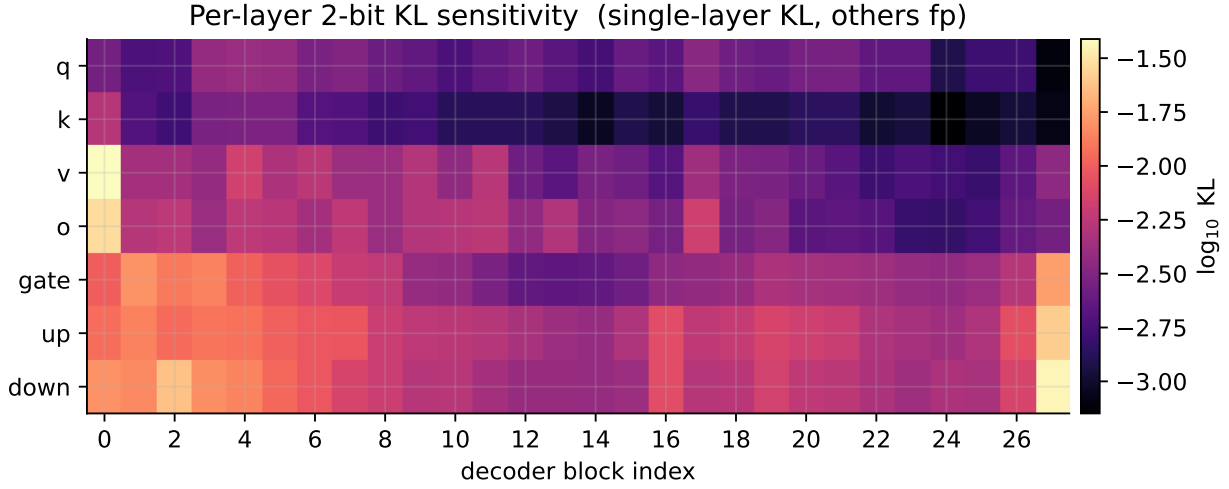


Figure 5: Per-layer 2-bit KL sensitivity map (single-layer KL, others full precision), $\log_{10}$ scale. MLP and value/output projections carry the sensitivity; query/key are nearly free.

Table 2: Per-layer allocation vs. uniform (TCQ+eq, Qwen3-0.6B). The 3.0-bit row is an equal-budget comparison.

Budget	Model	bits/w	KL	perplexity	layers (2/3/4 bit)
3.0 (equal)	Mixed (allocated)	3.023	0.238	34.76	24 / 149 / 23
	Uniform 3-bit	3.023	0.276	36.59	0 / 196 / 0
2.5	Mixed (allocated)	2.523	0.494	44.79	101 / 91 / 4
–	Uniform 2-bit	2.023	1.386	105.18	196 / 0 / 0

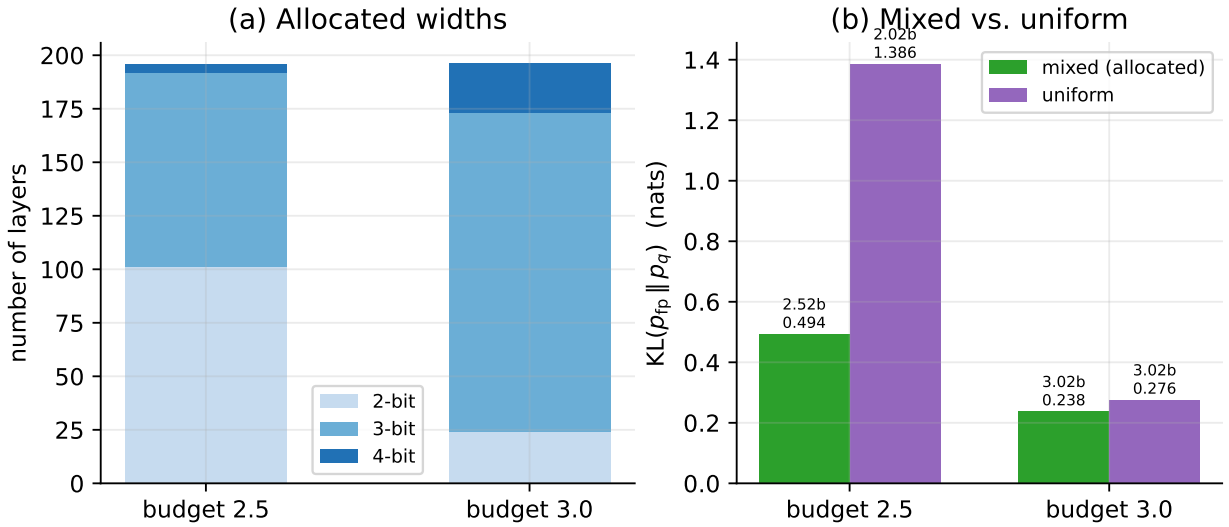


Figure 6: (a) Widths chosen by the allocator at each budget. (b) KL of mixed vs. uniform; annotations give bits/weight and KL. At equal 3.0 bits the mixed model wins.

5.6 The QJL-for-weights negative result

At an equal ~ 3 -bit budget, the unbiased QJL correction is markedly worse than plain nearest 3-bit. In a 16-layer synthetic chain (Fig. 7), nearest-3b reaches KL 0.096 / top-1 0.98 while 2b+QJL($k=d$, 3.06 bits) reaches KL 2.58 / top-1 0.01; on a real 135M model at equal budget the QJL-corrected model (KL 9.75) is $\sim 5.7\times$ worse than nearest-3b (KL 1.72). The estimator debiases exactly as predicted, but its noise standard deviation $\sqrt{\pi/2} \|r\| \|x\| / \sqrt{k}$ exceeds the bias it removes even at $k=d$ (a full extra bit), and the variance compounds multiplicatively through depth. The random rotation already decorrelates residuals from activation structure, removing the failure mode the correction targets. QJL remains the right tool where TurboQuant proved it—KV-cache / attention inner products, where the error path is shallow.

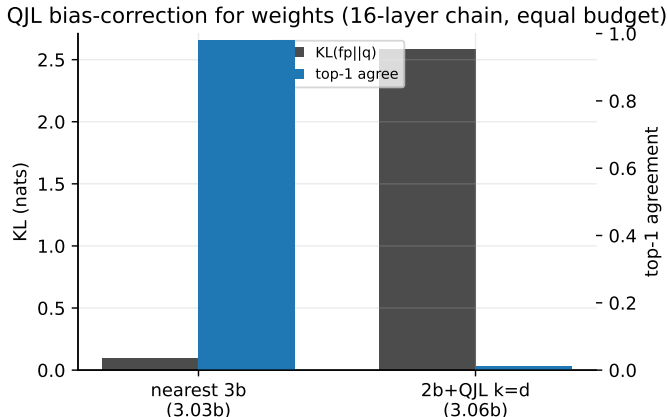


Figure 7: Unbiased QJL correction for *weights* loses to deterministic rounding at equal budget (16-layer chain). Correction variance > removed bias.

6 Discussion, Positioning, and Limitations

What is and is not novel. The pipeline shape—rotation, strong codebook, error feedback, mixed precision—overlaps substantially with recent work [6, 8, 16–18, 24]; we make no state-of-the-art claim. Our defensible analytical contributions are the quarter-power equilibration exponent (Proposition 1), which we position explicitly against BASE-Q’s $1/2$ (a different objective) and AWQ’s grid-tuned heuristic, and the QJL-for-weights negative result. The AM–GM / Cauchy–Schwarz *machinery* for optimal scaling is itself established [1, 6]; only the exponent for the weight-only, post-rotation regime is, to our knowledge, unstated elsewhere.

Limitations. (i) Scale: our evidence is at ≤ 0.6 B parameters; the 2-bit ranking in particular may shift at 7B+. (ii) Baselines: we compare against our own controlled configurations, not against released GPTQ/AWQ/QuIP#/QTIP checkpoints on WikiText-2 and zero-shot suites; those comparisons are required before any efficiency claim. (iii) The equilibration result rests on the isotropic-error and MSE-proportional-to-energy modeling assumptions of §3.1; it is an approximation, not an exact optimality theorem. (iv) Ours is a measurement implementation: dequantized weights are materialized in the model dtype rather than packed low-bit kernels, so numerics—but not latency—match a deployed kernel.

Next step. The data point directly at combining error feedback with the trellis (block-LDLQ over TCQ, in the spirit of QuIP#/QTIP), since trellis wins at 2 bits and error feedback is cheap and complementary; evaluated at 7B against standard baselines, that combination is the natural methods contribution.

7 Conclusion

TurboPress is a reproducible harness that measures, at matched bit budgets, the marginal value of each ingredient of modern rotated PTQ. Its measurements yield a corrected equilibration exponent, a clean negative result on unbiased weight correction, a demonstration that error feedback and trellis coding are complementary, and an equal-budget win for KL-driven mixed precision. We release the code, configurations, and exact result logs.

Reproducibility. All experiments are seeded and deterministic on a given device; the release includes the quantization modules, the sweep and allocation drivers, 89 unit tests (exact identities, statistical bounds, trellis round-trip, LDLQ equivalence, allocation properties), and the raw result JSON from which every table and figure is generated.

References

- [1] Juan Amboage, Pablo Monteagudo-Lago, Ian Colbert, Giuseppe Franco, and Nicholas Fraser. Optimal post-training quantization scales and where to find them. *arXiv preprint arXiv:2606.10890*, 2026.
- [2] Saleh Ashkboos, Amirkeivan Mohtashami, Maximilian L. Croci, Bo Li, Martin Jaggi, Dan Alistarh, Torsten Hoefer, and James Hensman. QuaRot: Outlier-free 4-bit inference in rotated llms. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2404.00456.
- [3] Jerry Chee, Yaohui Cai, Volodymyr Kuleshov, and Christopher De Sa. QuIP: 2-bit quantization of large language models with guarantees. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2307.13304.
- [4] Elias Frantar and Dan Alistarh. Optimal brain compression: A framework for accurate post-training quantization and pruning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2208.11580.
- [5] Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2210.17323.
- [6] Liulu He, Shenli Zheng, Karwei Sun, Yijiang Liu, Yufei Zhao, Chongkang Tan, Huanrui Yang, Yuan Du, and Li Du. BASE-Q: Bias and asymmetric scaling enhanced rotational quantization for large language models. *arXiv preprint arXiv:2506.15689*, 2025.
- [7] Hugging Face TB. SmolLM2: When smol goes big, 2025. Model: HuggingFaceTB/SmolLM2-135M, Hugging Face Hub.

- [8] Donghyun Lee, Jitesh Chavan, Duy Nguyen, Sam Huang, Liming Jiang, Priyadarshini Panda, Timo Mertens, and Saurabh Shukla. OrbitQuant: Data-agnostic quantization for image and video diffusion transformers. *arXiv preprint arXiv:2607.02461*, 2026.
- [9] Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Xingyu Dang, and Song Han. AWQ: Activation-aware weight quantization for on-device llm compression and acceleration. In *Proceedings of Machine Learning and Systems (MLSys)*, 2024. arXiv:2306.00978.
- [10] Zechun Liu, Changsheng Zhao, Igor Fedorov, Bilge Soran, Dhruv Choudhary, Raghuraman Krishnamoorthi, Vikas Chandra, Yuandong Tian, and Tijmen Blankevoort. SpinQuant: Llm quantization with learned rotations. *arXiv preprint arXiv:2405.16406*, 2024.
- [11] Stuart P. Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28(2):129–137, 1982.
- [12] Michael W. Marcellin and Thomas R. Fischer. Trellis coded quantization of memoryless and gauss-markov sources. *IEEE Transactions on Communications*, 38(1):82–93, 1990.
- [13] Joel Max. Quantizing for minimum distortion. *IRE Transactions on Information Theory*, 6(1):7–12, 1960.
- [14] Markus Nagel, Mart van Baalen, Tijmen Blankevoort, and Max Welling. Data-free quantization through weight equalization and bias correction. In *International Conference on Computer Vision (ICCV)*, 2019. arXiv:1906.04721.
- [15] Qwen Team. Qwen3 technical report, 2025. Model: Qwen/Qwen3-0.6B, Hugging Face Hub.
- [16] Preetam Sharma and Kacper Dobek. QAM-W: Joint 2d codebook quantization for llm weights via hadamard rotation and activation-aware scaling. *arXiv preprint arXiv:2605.26339*, 2026.
- [17] Albert Tseng, Jerry Chee, Qingyao Sun, Volodymyr Kuleshov, and Christopher De Sa. QuIP#: Even better llm quantization with hadamard incoherence and lattice codebooks. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.04396.
- [18] Albert Tseng, Qingyao Sun, David Hou, and Christopher De Sa. QTIP: Quantization with trellises and incoherence processing. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2406.11235.
- [19] Gottfried Ungerboeck. Channel coding with multilevel/phase signals. *IEEE Transactions on Information Theory*, 28(1):55–67, 1982.
- [20] Tycho F. A. van der Ouderaa, Maximilian L. Croci, Agrin Hilmkil, and James Hensman. Pyramid vector quantization for llms. *arXiv preprint arXiv:2410.16926*, 2024.
- [21] Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. SmoothQuant: Accurate and efficient post-training quantization for large language models. In *International Conference on Machine Learning (ICML)*, 2023. arXiv:2211.10438.
- [22] Amir Zandieh, Majid Daliri, and Insu Han. QJL: 1-bit quantized jl transform for kv cache quantization with zero overhead. *arXiv preprint arXiv:2406.03482*, 2024.
- [23] Amir Zandieh, Majid Daliri, Majid Hadian, and Vahab Mirrokni. TurboQuant: Online vector quantization with near-optimal distortion rate. *arXiv preprint arXiv:2504.19874*, 2025.

- [24] Shihao Zhang, Haoyu Zhang, Ian Colbert, and Rayan Saab. Qronos: Correcting the past by shaping the future in post-training quantization. *arXiv preprint arXiv:2505.11695*, 2025.